BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: MRD-XII-21 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes	**Maximum Marks:** 20	**Paper Code:** MRD-XII-21 (A) VERSION: A

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A,B,C, or D.

Q1. If $10^x = 2$, then x = ...

(A) log 10
(B) log 2
(C) log 1
(D) log10/log2

Q2. The function $f(x) = (x^2-4)/(x-2)$ is not continuous at x = ...

(A) 2
(B) −2
(C) 4
(D) None of these

Q3. $d/dx \sin^{-1}x$ = ...

(A) $1/\sqrt{(1-x^2)}$
(B) $\pm 1/\sqrt{(x^2-1)}$
(C) $1/(1+x^2)$
(D) $1/(x\sqrt{(x^2-1)})$

Q4. The derivative of $f(x) = x^2$ at a point P(3, 9) is equal to .....

(A) 18
(B) 27
(C) 6
(D) Zero

Q5. If h(x) = f(x)/g(x), g(x) ≠ 0 and f(x), g(x) are differentiable functions, then by Quotient rules h'(x) =

(A) $(f'(x)g(x) - g'(x)f(x))/(f(x))^2$

(B) $(f'(x)g(x) - g'(x)f(x))/(g(x))^2$

(C) $(f'(x)g(x) - g'(x)f(x))/$[illegible in source scan]

(D) $(f'(x)g(x) + g'(x)f(x))/(f(x))^2$

Q6. If $y = 3x^3 + 4x + 5$ then $y'' =$ is .....

(A) 27

(B) 18x+4

(C) 18x

(D) 27x

Q7. If f(x) = tan x, then f'(0) = ...

(A) Zero

(B) 1

(C) 2

(D) 1

Q8. If F(t) = [illegible in source scan] is a vector function, then $(e^tF)(t)$ = ...

(A) [illegible in source scan]

(B) $e^tt^2 + e^tt + t\sin t$

(C) e^t

(D) $e^tt^2 + e^tj - e^{-\sin t}k$

Q9. $\int e^x\,dx$ = ...

(A) 1

(B) e

(C) 1−e

(D) e−1

Q10. $\int 4x^{-8}\,dx$ = ...

(A) $-32x^{-9} + c$

(B) $-4/7x^{-7} + c$

(C) $-4/7x^{-7} + c$

(D) $-4/8x^{-9} + c$

Q11. $1/2 \int d\theta$ = ...

(A) $\theta/2 + c$

(B) $\theta + c$

(C) $x/2 + c$

(D) Zero

Q12. y − intercept of $y = -3x + 9$ is equal to .....

(A) 3

(B) −3

(C) 9

(D) −9

Q13. Slope of a straight line which is parallel to $2y - 4x = 7$ is equal to .....

(A) 1/2

(B) 2

(C) −2

(D) −1/2

Q14. The co-ordinates of the centre c(−g, −f) of the circle $x^2 + y^2 - 8x - 6y + 9 = 0$ are .....

(A) (4,3)

(B) (8,6)

(C) (−4,−3)

(D) (−8,−6)

Q15. Equation of the circle having center (0, 0) and radius 4 units is given by .....

(A) $x^2 + y^2 = 4$ (B) $x^2 + y^2 = 16$

(C) $x^2 + y^2 = 2$ (D) $x^2 + y^2 + 4 = 0$

Q16. Which one of the following parabolas opens downward?

(A) $x^2 = 8y$ (B) $y^2 = 8x$

(C) $y^2 = -8x$ (D) $x^2 = -8y$

Q17. In standard form, the hyperbola $16y^2 - 9x^2 = 144$ can be written as .....

(A) $x^2/4 + y^2/9 = 1$ (B) $y^2/9 - x^2/16 = 1$

(C) $x^2/9 + y^2/4 = 1$ (D) $y^2/9 + x^2/16 = 1$

Q18. Which one is homogeneous differential equation?

(A) $dy/dx = (x+y)/(x-y)$ (B) $(x-y)\, dy/dx = x^2 + y^2$

(C) $dy/dx = x^2 - xy$ (D) None of these

Q19. Domain of f(x,y) = e^((x+1)/(y−2)) is.....

(A) $x,y \in R$ and $y \neq -2$ (B) $x,y \in R$ and $x \neq 2$

(C) $x,y \in R$ and $y \neq 2$ (D) $x,y \in R$ and $x \neq -2$

Q20. $f(x,y) = ax^2 + 2bxy + cy^2$ is homogeneous function of degree .....

(A) 2 (B) 1

(C) 0 (D) 4

Section-B

Maximum Marks: 50

Q2. Attempt any ten (10) of the following parts. All parts carry equal marks.

(i) Evaluate: Limit_{x→∞} ((3x²+2)/(x²+1))

(ii) Differentiate the implicit equation $(x + y)^3 + 3y = 3$

(iii) Find the equation of tangent line to the curve $y = \sin(2x + \pi)$ at $x = 0$

(iv) If $v = 2i - j + 5k$ and $w = i + 2j - 3k$, then find $d/dt\ (v + tw)$

(v) Integrate $\int xdx/\sqrt{(x^2-4)}$

(vi) Show that the lines $x - y - 2 = 0$, $2x - y - 5 = 0$ and $11x - 5y - 28 = 0$ are concurrent and find the point of concurrency.

(vii) Find the center C (−g, −f) and radius (r) of the circle $4x^2 + 4y^2 + 16x - 12y - 7 = 0$

(viii) For what value of 'c' the line $x - y + c = 0$ will touch the parabola $y^2 = 8x$.

(ix) Find the equation of the tangent at a point P(3, 12/5) to ellipse $x^2/25 + y^2/9 = 1$

(x) Solve the initial value problem $dy/dx = 4xy$, $y(3) = 1$

(xi) Find the partial derivatives of the function $f(x, y) = xy^3 \tan^{-1} y$.

(xii) Evaluate: Limit_{x→a} (x − a)/(√x − √a)

(xiii) Find dy/dx, if y = xe^{x ln x}

Section-C

Maximum Marks: 30

> Note: Attempt any three (3) questions. All questions carry equal marks.

Q3.

(a) Use Maclaurin's series expansion for e^x and show that the sum of 1 + 1/1! + 1/2! + 1/3! + ... is e.

(b) Integrate $\int x^2 \ln x \, dx$

Q4.

(a) Find the angle from the line 7x + 3y + 9 = 0 to the line 5x − 2y + 2 = 0

(b) Solve the homogeneous differential equation $dy/dx = (xy - y^2)/x^2$

Q5.

(a) Use any suitable rule of differentiation to perform dy/dx for y = [illegible in source scan]

(b) Find the equation of the circle which contains the point (2, 6), (6, 4) and has its center on the line 3x + 2y − 1 = 0

Q6.

(a) Verify Euler's theorem for $z = f(x, y) = ax^2 + 2hxy + by^2$

(b) Find the equation of hyperbola with vertices at (2, −2), (−4, −2) and that passes through P(5, 1).